OWL2 EXERCISES: TABLEAU REASONING

Ex1. Consistency checking
Given an Ontology
:Penguin rdfs:subClassOf :Bird .
:Bird rdfs:subClassOf :Animal .
:Animal owl:disjointWith :Plant .
:Pingu a :Penguin , :Plant .
Convert to NNF, run the tableau algorithm, and decide whether the ontology is consistent.

Ex2. Class instantiation
Given an Ontology
```
:Parent owl:equivalentClass [
   owl:intersectionOf ( :Person
      [ a owl:Restriction ; owl:onProperty :hasChild ;
        owl:someValuesFrom :Person ] ) ] .
:Mary a :Person ; :hasChild :Tom .
```

a) Can the reasoner derive that Mary is a Parent?
b) If we add ":Tom a :Person .", does the result change? Explain the difference.

Ex3. Subclass relation with quantifiers
Given an ontology in Description Logic notation:
1. Professor ⊑ ∃supervises.PhDStudent
2. PhDStudent ⊑ Student ⊓ Researcher
3. Professor ⊑ ∀supervises.(¬Student ⊔ Employee)

a) Prove or disprove: Professor ⊑ ∃supervises.Employee (every professor supervises at least one employee).
b) If axiom 3 is removed, how does the conclusion change?